TT405 / MS300 — Label and Barcode Guide

TT405 and MS300 send the label of each cut piece directly to the label printer in the **EPL2** command language (serial port or USB). Both machines use the same command structure; the text lines and the barcode content differ per machine. This page shows the commands sent to the printer for a cut list row and which fields make up the barcode.

Command sequence

For every label the commands are sent in the following order. Each line ends with CR LF. Example values are the default settings.

Command	Example	Meaning	Machine setting (Printer Settings)
(LF)		TT405 only: a single line feed	—
`N`	`N`	Clears the printer image buffer	—
`Y`	`Y38,N,8,1`	Serial port: 38400 baud, no parity, 8 bits, 1 stop	Fixed
`Q`	`Q160,24,`	Label length and gap between labels (dots)	Label → Label Length, Gap Length
`q`	`q840`	Label width (dots)	Label → Label Width
`R`	`R5,1`	Reference point offset (horizontal, vertical)	Label → Horiz. / Vert. margin
`S`	`S3`	Print speed	Label → Speed
`OD`	`OD`	Direct thermal printing	Fixed
`D`	`D15`	Density. On MS300 V26.1.11.8 and earlier this value is sent as a second `S` line instead of `D` (`S10`).	Label → Density
`A`	`A30,50,0,2,1,1,N,"PROG NO:1"`	Text line (one per enabled line)	Text tab
`B`	`B30,110,0,1B,2,2,40,N,"…"`	Barcode	Barcode tab
`B`	`B30,0,0,1B,2,2,40,N,"0001"`	TT405 only, **if the 2nd barcode is active**: cut number	2. Barcode Active
`Z`	`ZB`	TT405 only: print direction (`T` / `B`)	Label → Direction
`P`	`P1`	Print one label	—

1 dot = 0.125 mm (203 dpi). `q840` is about 105 mm, `Q160` about 20 mm.

Text line (`A`) parameters

`A` horizontal, vertical, rotation, font, horizontal multiplier, vertical multiplier, normal/reverse, `"text"`

Order	Parameter	Values
1	Horizontal start	dots
2	Vertical start	dots
3	Rotation	0 normal · 1 = 90° · 2 = 180° · 3 = 270°
4	Font	1 (smallest) … 5 (largest)
5–6	Horizontal / vertical multiplier	1, 2, …
7	Appearance	`N` normal · `R` reverse (white on black)

Barcode (`B`) parameters

`B` horizontal, vertical, rotation, type, narrow bar width, wide bar width, height, human readable, `"data"`

The default type is `1B` (Code 128 B). With human readable `B` the barcode data is also printed below the barcode; with `N` it is not.

Text lines

Label text contains no Turkish characters. Line numbers match the order on the machine's Printer Settings → Text tab.

No	Name on settings page	TT405 (Turkish)	TT405 (English)	MS300	Source field
0	Logo	company name	company name	company name	Fixed
1	Prog No	`PROG NO:`	`PROG NO:`	`PROG NO:`	1 Cut Number
2	Car No	`ARABA NO:`	`CAR NO:`	`ARABA NO:`	11 Cart No (TT405: last digit)
3	Operation	operation name	operation name	operation name	15 Operation
4	Loc. on frame	`UST(Y) …`	`UP(Y) …`	`UST(Y) …`	10 Assembly
5	Location	`YER NO:`	`LOC NO:`	`YER:`	12 Location No (TT405: last 3 digits)
6	Poz No	`POZ NO:`	`POZ NO:`	`POZ NO:`	9 Position No
7	Length	`UZUNLUK: … mm`	`VALUE: …mm`	`LENGTH: … mm`	2 Cut Length
8	Frame	`CERCEVE:(X x Y)`	`FRAME:(X x Y)`	`CERCEVE:(X x Y)`	2 and 13, weld allowance subtracted
9	Profile	`PROFIL:`	`PROFIL:`	`PROFIL:`	3 Profile Code (max. 12)
10	Dealer	`BAYI:`	`VENDOR:`	`BAYI:`	16 Dealer Name (TT405: 16, MS300: 12 characters)
11	Customer	`MUSTERI:`	`CLIENT:`	`MUSTER:`	8 Order No (10 characters)

No	Name on settings page	TT405 (Turkish)	TT405 (English)	MS300	Source field
14	TT405: Height · MS300: Angle	`YUKSEKLIK:`	`HEIGHT:`	`ACI:45/90`	TT405: printed only with barcode design 2 · MS300 (V26.1 and later): 4 Left Angle / 5 Right Angle

X and Y on the frame line are the length and other axis values minus the **weld allowance** (Settings page).

MS300 angle line (PLC software V26.1 and later): the left / right angles from the cut list are printed as a separate text line: `ACI:45/90`. It is selected as **Angle** (14) in the Printer Settings → Text list; its position, font, rotation and enable state are set like the other lines. If both angles are 0 the line is not printed. The angles are not part of the barcode data. Software before V26.1 has no angle line; choose PLC software → Before V26.1 in the simulator to hide it.

Lines that change with TT405 barcode design 2

No	Turkish	English	Value
1	`SIPARIS NO:`	`ORDER NO:`	Cut Number
5	`KALIP FARKI: ...mm`	`MOLD DIFFERENCE: ...mm`	Location No, last 2 digits
6	`MODUL NO:`	`MODULE NO:`	Position No
8	`DIGER UZUNLUK: ...mm`	`OTHER VALUE: ...mm`	Other Axis
14	`PROFIL KALINLIGI: ...mm`	`PROFILE WIDTH: ...mm`	Height (max. 3 digits)

With barcode design 3, line 5 becomes `BIN NO:` in English.

Barcode content

The barcode data is the fields joined one after another; there is no separator. Numeric fields are padded with leading `0`. Examples use sample row 1 of the cut list guide (`1;806;XXXXXXX12PEN;45;45;1;60;X10MUSTERI;1;1;1;1;906;11;0;…`) with a weld allowance of 0.

TT405 — Barcode design no 0 (Standard barcode)

Order	Digits	Content
1	1	`K` (fixed)
2	1	Cart No, last digit
3	3	Location No
4	12	Profile Code, padded with spaces on the right to 12 characters
5	2	Colour / Gasket
6	6	Length − weld allowance; 4 whole + 2 decimal digits (0.01 mm)
7	6	Other Axis − weld allowance; 4 whole + 2 decimal digits

Example: `K1001XXXXXXX12PEN11080600090600` (31 characters)

TT405 — Barcode design no 1

Order	Digits	Content
1	1	K (fixed)
2	4	Cut Number
3	2	Position No, last 2 digits
4	1	Cart No, last digit
5	2	Location No, last 2 digits
6	2	`00` — colour field; always `00` in the current software
7	5	Order No, last 5 characters

Example: `K00010110100STERI`

TT405 — Barcode design no 2

Order	Digits	Content
1	4	Length, whole mm (weld allowance not subtracted)
2	4	Other Axis, whole mm
3	2	Location No, last 2 digits
4	3	Height

Example: `08060906` + `01` + `060` = `0806090601060`

TT405 — Barcode design no 3

The barcode contains only the **Dealer Name** (max. 20 characters).

TT405 — Barcode design no 4

Order	Digits	Content
1	10	Profile Code, first 10 characters
2	5	Length; 4 whole + 1 decimal digit (0.1 mm)
3	5	Other Axis; 4 whole + 1 decimal digit

Example: `XXXXXXX12P0806009060`

TT405 — 2nd barcode

If **2. Barcode Active** is selected in Printer Settings, a second barcode is printed after the main barcode. It contains the 4-digit cut number (e.g. `0001`). Its position is kept separately; the default is the top left corner.

MS300

Order	Digits	Content
1	5	K1111 (fixed)
2	≤ 12	Profile Code (not padded)
3	2	Colour / Gasket
4	6	Length − weld allowance; 4 whole + 2 decimal digits
5	6	**Height** − weld allowance; 4 whole + 2 decimal digits

Example: `K1111XXXXXXX12PEN11080600006000`

The last field of the MS300 barcode is not the other axis as on TT405 but **Height**, field 7 of the cut list.

Default positions

The **Load SW Format** button on the Printer Settings page loads the positions below. Values are horizontal, vertical start in dots; all use font 2, multiplier 1 × 1, rotation 0 unless stated. Positions changed on site are stored on the machine.

No	Line	TT405 103x20	TT405 103x25	MS300
0	Logo (font 1, 90°, reverse)	20, 10	45, 40	10, 5
1	Prog No	30, 50	60, 30	45, 10
2	Car No	215, 50	225, 30	205, 10
3	Operation	660, 70	60, 50	45, 26
4	Loc. on frame	575, 70	400, 50	205, 26
5	Location	375, 50	225, 50	380, 10
6	Poz No (vertical multiplier 2)	650, 5	650, 5	650, 10
7	Length	30, 70	60, 70	45, 46
8	Frame	285, 70	315, 70	300, 46
9	Profile	30, 90	60, 90	45, 66
10	Dealer	285, 90	315, 90	300, 66
11	Customer	550, 90	600, 90	550, 66
14	Angle (MS300 only, V26.1 and later)	—	—	550, 46
—	Barcode (height)	30, 110 (40)	60, 110 (60)	30, 90 (40)
—	2nd barcode	30, 0 (40)	30, 0 (40)	—

Label settings (all): length 160, gap 24, width 840, margin 5 / 1, speed 3, direction B. Density is 15 on TT405 and 10 on MS300.

If the preset is loaded while TT405 barcode design 2 is selected, these positions change: Prog No 30, 5 (font 3) · Car No 30, 86 · Location 30, 26 · Poz No 575, 26 · Customer 575, 66 · Height 285, 26 · barcode 250, 84 (human readable on).

Known behaviour

- **MS300 density (V26.1.11.8 and earlier):** the density value is sent with a second `S` (speed) command instead of `D`; the printer may interpret it as speed. Later versions send `D`; the simulator shows `D`.
- **TT405 barcode design 1:** the colour field is always `00`; the colour code from the cut list is not used in this design.
- **Poz No line:** loading the preset does not change this line's rotation; the previous value is kept.
- **MS300 C70 panel, Barcode tab:** the vertical start box is labelled "Narrow bar width" and the narrow bar width box is labelled "Vert. start" (labels swapped). The simulator uses the correct names and shows the panel label as a note.
- **Label length:** in the 103x25 layout the bottom of the barcode exceeds the 160-dot label length; set the label length accordingly.